MIL-M-12652A (Ord)
24 June 1954
SUPERSEDING
MIL-M-12652 (Ord)
21 April 1953

MILITARY SPECIFICATION
MOTOR, D-C, 24 VOLTS, 1-1/2 H.P.

1. SCOPE

1.1 This specification covers a motor which is a component of Gear, Oil, M6A1B1, and Gear, Oil, M6A1E1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS
Military

MIL-S-10379	Suppression, Radio Interference, General Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-P-12589	Packaging of Motor DC 1½ H. P.

U. S. Army

51-7	Instruments, Fire Control, General Specification Governing the Manufacture and Inspection of

DRAWINGS
Ordnance Corps

F7666417	Motor, D-C

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

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3.1 General. - Unless specificed otherwise, the requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory.

3.2 Material. - Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Electrical

3.3.1 Balance. - The armature shall be in dynamic balance at all speeds up to and including 8000 rpm. The balancing shall be accomplished by such means as will exclude the possibility of balancing material shifting and destroying armature balance. Solder shall not be used as a method of balance.

3.3.2 Brush position. - The brushes shall be set at the optimum position for desired rotation and maximum efficiency; the position shall be inscribed properly.

3.3.3 Output rating. - The motor shall be capable of developing 1-1/2 H.P. for a period of one hour, and 3 H.P. for a period of 15 seconds, when 24 volts d-c is applied to the input terminals. The motor shall not be damaged when 30 volts d-c is applied for a period of 15 minutes while the motor is operating with full load.

3.3.4 Speed. - The speed of the motor at rated load and voltage shall be not less than 4000 rpm when the motor is energized initially; the speed after 30 minutes under the same conditions shall not exceed 4425 rpm.

3.3.5 Operating current. - The motor shall be operated with rated input voltage at no load and at full load. The rated input current between motors shall vary not more than 10 percent at no load and 5 percent at rated load.

3.3.6 High potential. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure of the armature assembly, brush holder assembly, and frame and field assembly when subjected to the test specified in 4.2.1.

3.3.7 Insulation resistance. - Each armature assembly, brush holder assembly, and frame and field assembly shall have a minimum insulation resistance of 10 megohms when subjected to the test specified in 4.2.2.

3.3.8 Temperature rise. - The temperature rise of the motor shall not exceed 80°C(144°F) after it has been subjected to one hour of continous operation at rated load. Test shall be conducted in accordance with 4.2.3.

3.3.9 Radio interference suppression. - The motor shall meet the requirements of Specification MIL-S-10379 for radio interference suppression.

3.4 Environmental Tests

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3.4.1 Temperature. - The motor shall meet the requirement of 3.3.3 throughout the air temperature range extending from plus 51.5°C (≠ 125°F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr), to minus 31.2°C (- 25°F) (minimum exposure of 3 days without benefit of solar radiation). The motor shall be capable of safe storage within the air temperature range extending from plus 71°C (≠ 160°F) (for periods as long as 4 hours per day) to minus 62.2°C (- 80°F) (for periods of at least 3 days duration). Test shall be conducted in accordance with 4.3.2.

3.4.2 Humidity. - The motor shall meet the requirements of 3.3.3 immediately after being subjected to the humidity test specified in 4.3.3.

3.4.3 Rain. - There shall be no evidence of water penetration or damage to the motor when subjected to the rain test in accordance with 4.3.4.

3.4.4 Vibration. - The motor shall not suffer damage or fail to meet the requirements of 3.3.3 immediately after being subjected to the vibration test specified in 4.3.5.

3.4.5 Shock. - The motor shall not suffer damage or fail to meet the requirements of 3.3.3 immediately after being subjected to the shock test specified in 4.3.6.

3.5 Workmanship. - Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General. - Unless otherwise specified, each motor shall be inspected for conformance with this specification and Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60°F to 90°F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Electrical

4.2.1 High potential. - Components specified shall be subjected to a potential of 900 volts rms at a frequency of 60 cycles per second for a period of one minute. Three of the first five and one out of every additional 50 or fraction thereof shall be subjected to the test voltage, which shall be applied at a rate not exceeding 500 volts rms per second. For all other units the test voltage shall be applied for a period not less than 5 seconds or more than 30 seconds. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance. - The insulation resistance of components specified shall be measured by an approved method using a source of d-c potential of 500 volts.

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4.2.3 Temperature rise. - Three of the first five motors and one out of every additional 50 or fraction thereof shall be tested. The motor shall be tested in still air and shall not be mounted in a manner that will cause heat dissipation through the mounting. The temperature rise shall be determined as follows:

a. A stop watch shall be started at the instant the unit is stopped.

b. Necessary bridge connections shall be made.

c. A value of resistance which is slightly smaller than the resistance of the winding shall be preset into the bridge. Observe the null indicator of the bridge; the instant the needle crosses the "zero" position, record the time and the value of the resistance set into the bridge. This procedure shall be followed until five values of resistance versus time after shutdown are recorded. The resistance values shall then be plotted against the corresponding time, and the curve extrapolated to zero time. The value of "R" at zero time shall be used in equation (2) in order to obtain "T". The value "T" shall then be used in equation (1) to obtain "T_r".

(1) $T_r = T - T_a$

(2) $T = \frac{R}{r} (234.5 + t) - 234.5$

Where: T_r = Temperature rise in degrees centigrade.

T = Winding temperature in degrees centigrade

T_a = Ambient temperature in degrees centigrade at end of temperature rise test.

R = Hot resistance of winding in ohms.

r = Cold resistance of winding in ohms.

t = Temperature of winding in degrees centigrade when cold resistance measurement was made.

4.2.4 Radio interference. - The motor shall be tested for suppression of radio interference in accordance with Specification MIL-S-10379.

4.3 Environmental tests

4.3.1 General. - The environmental tests specified shall be limited to those contained in the following paragraphs. Three of the first five motors and one out of every additional 100 or fraction thereof shall be tested.

NOTE: These tests shall be eliminated if the motor is tested as a component of the oil gear assembly, in which case it shall conform with the environmental requirements of the oil gear specification.

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4.3.2 Temperature. - The motor shall be exposed to an air temperature of minus 62.2°C (-80°F) for a period of 3 days. The air temperature shall then be raised to minus 31.2°C (-25°F) and after the internal temperature of the unit has reached equilibrium (approximately 4 hours), it shall be tested for conformance with 3.3.3. The air temperature shall be raised to plus 71°C (≠160°F) and after the internal temperature of the unit has reached equilibrium at that temperature it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (≠125°F) and after the internal temperature of the unit has reached equilibrium at that temperature the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The unit then shall be tested for conformance with 3.3.3.

4.3.3 Humidity. - A test chamber that is capable of being sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the motor being tested. A unit that has been tested and found to be in conformance with the electrical requirements of this specification shall be placed in the chamber, the temperature of which has been adjusted to plus 46.1°C ≠ 1.1°C (≠ 115°F ≠ 2°F) and the relative humidity adjusted to 95 ≠ 5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be readjusted to plus 26.7°C ≠ 1.1°C (≠ 80°F ≠ 2°F) and the unit shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test the unit shall be tested for conformance with 3.3.3.

4.3.4 Rain. - The motor shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C to 30°C (68°F to 86°F) throughout the test period. A simulated rainfall of 4 plus 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C to 24°C (51.8°F to 75.2°F). The water spray shall be directed on the motor for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfaces of the motor that are normally subjected to rain when the instrument is in its operating position.

4.3.5 Vibration. - The motor shall be mounted using its own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursion as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated; the rate of change shall not exceed one cycle per second.

Frequency limits	Excursion	Acceleration	Time (Each direction)
0 - 10 cps	0.060 inch	- - - - - - -	1 hour
10 - 20 cps	0.030 inch	- - - - - - -	2 hours
20 - 400 cps	- - - - -	2.5 g's	2 hours

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4.3.6 Shock. - The motor shall be mounted using its own mounting provisions to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured record of shock pulse.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with specification MIL-P-12589.

6. NOTES

6.2 Ordering data. - Contracts and invitations for bids should specify the title, number, date of this specification and type of packing.

NOTICE: This specification, together with specifications and drawings pertaining to it, and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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